# POLAR WINDOW

A lifecycle planner for CLPS lunar deliveries, from launch to return, with the uncertainty and source of every number

Submission report for the 2026 challenge *CLPS Lunar Mission Browser*. Engine checks, response times, test counts and screens were measured on commit `9d1d5b1` of branch `featured`.

**Abstract**

NASA's Commercial Lunar Payload Services (CLPS) programme buys the delivery of science payloads to the Moon from commercial providers. Planning one delivery draws on separate tools for launch design, weather, trajectory, radiation, communications and recovery, each with its own conventions. This report describes POLAR WINDOW, an open planner that follows one CLPS delivery through six sections: readiness, launch, trip, landing, contact with Earth and return. Every number it serves states its source, its margin or the reason it has none. It also carries the age of any live input. No section returns a verdict or a combined score. The engines are checked against the public record of flown missions. The references are NASA's reconstructed trajectories of four CLPS landers, their published liftoff times and NOAA's list of solar particle events. Measured lunar radiation doses and the landing point of the OSIRIS-REx capsule complete the set. Designed at the direction each vehicle climbed, the launch instants of the three flown Falcon 9 missions fall within one minute of the published liftoffs. One day of propagation from a reconstructed state stays within one kilometre in all but one case. Unmet planned tolerances are reported with their measured values. When the climb direction is not known in advance, a designed launch instant misses by about 16 to 18 minutes. The default NRLMSIS 2.1 atmosphere lands the capsule within 6.6 kilometres of the measured touchdown, satisfying the 10 kilometre plan tolerance. It was chosen by a rule written after both landing results were known, using mesospheric measurements only partly independent of the models. Multi-year weather reforecasts show positive Brier skill at Vandenberg but fail on reliability, while Cape Canaveral lacks sufficient events, leaving wind shares marked as not checked. Link budgets and the return trajectory from the Moon remain unbuilt because their inputs are not public.

## 1 Introduction

NASA's Commercial Lunar Payload Services (CLPS) programme buys the delivery of science payloads to the Moon from commercial providers [1]. Four CLPS landers have flown: Peregrine Mission One, IM-1, Blue Ghost Mission 1 and IM-2. Their reconstructed trajectories are public in NASA's SPICE archive [2]. The 2026 NASA Space Apps challenge *CLPS Lunar Mission Browser* asks for a tool to compare landing sites and dates [3]. The tool is meant for planners, educators and the public, and it should show the Sun and Earth against the local horizon.

The first version of POLAR WINDOW answered one question. At a south-polar site and over chosen dates, it reported how long the Sun and Earth stay above the site's own skyline. A delivery raises many more questions. A project manager must know when the vehicle can leave the pad and whether the weather rules allow it. The trip passes through sunlight, shadow and the radiation belts. After landing, the team needs the hours in which a ground station can hear the lander. The launch vehicle's booster comes back to a landing zone on land or at sea. Each payload must also be built, tested and delivered on time. Each answer comes from a different tool and data source, and each tool states its assumptions in its own way.

This report describes the extension of POLAR WINDOW into a planner that follows one CLPS delivery through six sections: *Ready*, *Launch*, *Trip*, *Landing*, *Contact* and *Return*. The work makes four contributions.

1. A single contract for every number. Each value states its provenance: computed, published, entered by a named person, or assumed. It carries a margin (a ±, a range or a multiplicative factor) or the reason it has none. It also carries the age of any live input. No section returns a verdict, a readiness percentage or a combined score.
2. Trajectory, launch, radiation and entry engines checked against flown missions. The references are NASA's reconstructions of the four CLPS trips and the measured landing of the OSIRIS-REx sample capsule.
3. A process design that separates network access, long computations and request handling. Every stored answer records the inputs and engine versions it came from. Section 5.10 measures its response times and its job queue under load.
4. A disclosure of every planned acceptance tolerance that the engines do not meet, with the measured value beside it.

The evaluation is bounded by public data. Only four CLPS trips have reconstructions, and none of the planned missions publishes a trajectory, a radio link budget or a return state. The engines are therefore checked on the flown cases, and the planned missions receive designed answers labelled as such. Two external references named in the plan could not be run: the General Mission Analysis Tool (GMAT) and the SPENVIS web service. Section 5 states what replaced each of them.

## 2 Related work

**Illumination and Earth visibility at the lunar poles.** Polar lighting has been mapped from Lunar Orbiter Laser Altimeter (LOLA) topography [4] by casting horizons and averaging Sun visibility over years [5]. Later work located persistently illuminated regions in camera images [6] and in higher-resolution terrain models [7]. These studies produce fractions over long periods on a fixed grid. They answer where the light is on average, not what happens at one landing point over one mission's dates. They also report no timing uncertainty per crossing. The surface engine of POLAR WINDOW computes the Sun and Earth from the Navigation and Ancillary Information Facility's SPICE system [8, 9] with the DE440 ephemeris [10]. It reports each duration with a margin derived from the terrain's vertical error.

**Trajectory and launch design.** High-fidelity mission design is available in open tools such as GMAT, which has a documented verification programme [11]. Its building blocks are standard. They include Lambert solvers for the initial transfer [12], numerical integration under point-mass and harmonic gravity [13], and lunar gravity fields from GRAIL [14]. These tools require a scripted mission and an analyst. They do not read NASA's CLPS reconstructions or relate a launch day to the lighting rule at the landing site. They also do not label an input as published or assumed. POLAR WINDOW does not compete with them on fidelity. It connects a small, checked subset of their methods to the landing site, the pad and the reconstructions.

**Launch weather and forecast verification.** Launch weather criteria are published per vehicle as rule sets. The Falcon 9 rules, for example, limit the wind at 162 ft and lightning near the pad [15]. Ensemble systems such as the Global Ensemble Forecast System (GEFS) give a sample of possible weather [16]. Lightning is observed from orbit by the Geostationary Lightning Mapper (GLM) [17, 18], and long wind records come from the MERRA-2 reanalysis [19]. The standard score for a probability forecast is the Brier score [20]. Public presentations of launch weather usually give one chance of a rule being broken. They rarely state the sample behind it or whether that chance has been checked against observations.

**Radiation screening.** Trapped belt fluxes are commonly taken from the AE8 and AP8 models [21, 22]. The IRBEM library[1] implements them, and SpacePy exposes them in Python [23]. Galactic cosmic rays (GCR) can be modelled with one solar modulation parameter [24]. Measured doses exist in lunar orbit from CRaTER [25] and on the far-side surface from Chang'e-4 [26]. Large solar particle events are described by published spectra [27]. Engineering tools such as OLTARIS compute dose behind shielding once a trajectory and a geometry are known [28]. They need an account and a finished design, so they do not serve early screening of several candidate trips.

**Contact planning.** The Deep Space Network (DSN) publishes each antenna's horizon mask and link parameters [29]. Delay-tolerant networks route messages over scheduled contacts with contact graph routing [30]. Neither source combines a station's mask with the terrain around a polar lander, which decides whether Earth is visible at all.

**Gap.** Each area has mature tools, but each answers one phase with its own conventions. None carries the provenance and uncertainty of a number from the pad to the landing site and back. POLAR WINDOW keeps both attached to every number across the six phases of a delivery. Its engines are deliberately simpler than the specialist tools above. Designed values are labelled as designed, and the radiation answers name the tools in which a design must be confirmed.

## 3 System design

### 3.1 Processes and data flow

The system is one Python code base run as three long-lived processes over PostgreSQL and Redis (fig. 1). The *API* process answers requests from the browser and from the agents. It reads stored answers and feed snapshots, and it never waits for a long computation. The *compute worker* runs trajectory, launch-calendar and contact jobs. The *feeds worker* is the only process that reaches external services. It fetches weather, space weather, launch schedules and DSN activity, and it stores each response as a snapshot with its fetch time. An import contract enforces this rule in code. No module in the API, the services, the agents or the numerical core can import the feed package.

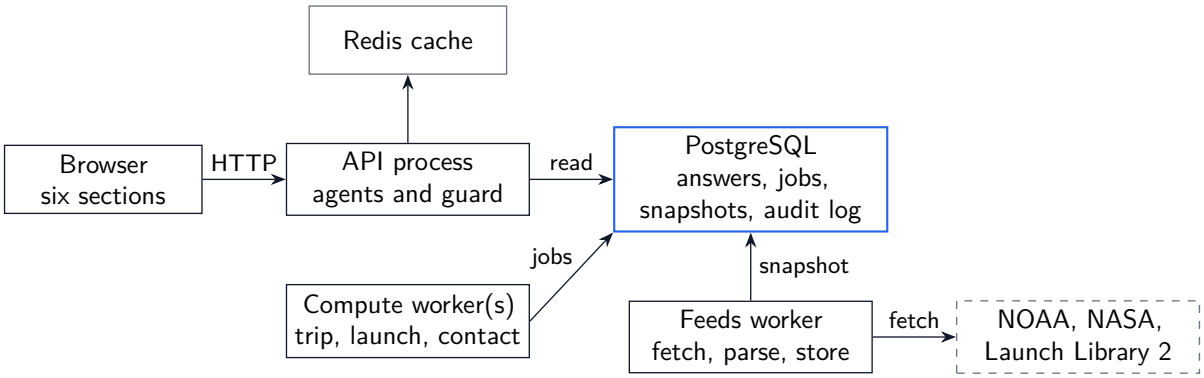



**Figure 1:** Processes. Only the feeds worker has outbound network access; the API queues work and reads stored answers.

An answer that takes seconds or minutes to compute is a *job*. Its key is a hash of the job kind, its parameters and the versions of every engine it depends on, called its *closure*. When a stored answer with that key exists, the API returns it at once. Otherwise the API queues a job and returns HTTP status 202 with a job reference, and the page polls until the answer is ready. Workers lease jobs from a PostgreSQL table with row locks that skip rows already taken.

[1] `https://prbem.github.io/IRBEM/`

Several workers can therefore run without a separate queue service. Each answer is stored as an *assessment* together with its inputs. These are the feed snapshot identifiers, the catalogue versions, the kernel set, the terrain product and any values a person entered. An answer is therefore reused until one of its inputs or engine versions changes.

### 3.2 One contract for every number

Every value that leaves the service has one shape, called a *quantity*. It holds the value, the unit, the provenance and exactly one form of margin: a symmetric ±, a low-to-high range, or a multiplicative factor. A value without a margin carries a sentence that says why. A value that cannot be given is null and carries its *missing reason*, for example that a vehicle's limit is not published. A duration cut short by the chosen dates is flagged as a lower bound. Probabilities use a second shape, called a *share*. It holds $k$ of $n$ model runs, an exact binomial interval [31] and a flag stating whether the share has been checked against observations.

Four rules of the project govern these shapes. No section returns a verdict: there is no go or no-go, no readiness percentage, no risk score and no overall chance of launch. Every number states whether it was computed, published, entered or assumed. Every live number shows its age, and past its staleness budget it is replaced by the reason there is no number. Every probability states its sample and its verification status. A schema test and the agents' guard check the first rule (section 4.9).

### 3.3 Layering, versions and catalogues

The numerical core has no input or output, no clock and no database access. Ten import contracts keep the layers apart. Among them, SPICE is reachable from one module only. Routes reach the domain only through services, and agents reach it only through their tool module. Each engine carries its own version, for example `traj-4` for the trajectory engine and `lw-5` for launch design. A change to one engine invalidates only the answers whose closure contains it.

Static facts live in nine cited catalogues: launch pads, vehicles, ground stations, relays, recovery zones, landing attempts, lessons, risk scales and weather rules. Every record carries its sources and compile date. A value that no source publishes is left empty with its reason and is never filled with a typical figure. The mission list itself holds 20 CLPS deliveries and 64 payloads, each payload with a source address.

## 4 Methods by mission section

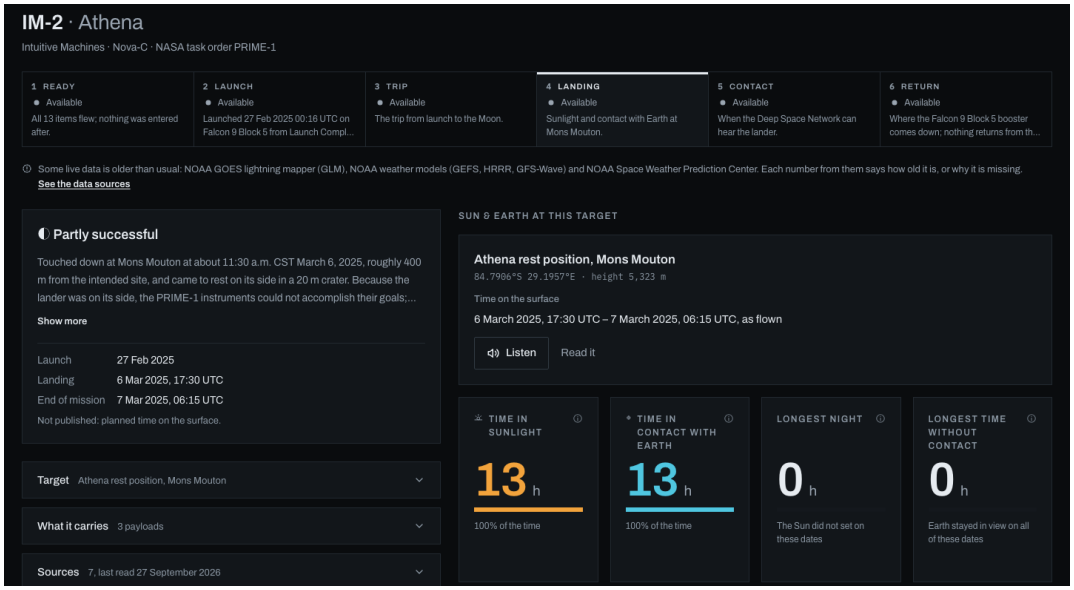
Table 1 lists the six sections, the question each one answers and the reference against which its engine is checked. The subsections below describe the engines in the order of a delivery. Figure 2 shows two sections as the project manager sees them. Each number carries its margin, or a sentence stating why it has none.

### 4.1 Landing: the Sun and Earth above the local skyline

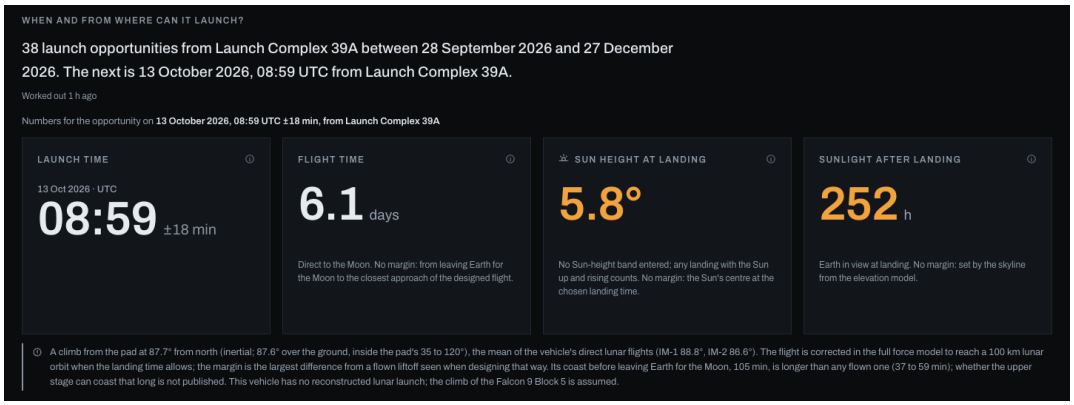
Terrain comes from the LOLA polar elevation models at 20 m (to 20 km from the site) and 80 m (to 50 km). Rays are cast on 1,440 bearings with the Moon's curvature removed, giving the elevation of the skyline in every direction. The Sun and Earth come from SPICE in the `MOON_ME` frame with light-time and stellar aberration corrections. A body is *up* when its centre clears the skyline at its azimuth. Each skyline crossing is located by bisection between time samples, so its time is not rounded to the step. Four durations follow over the chosen dates. The screens name them time in sunlight, time in contact with Earth, the longest night and the longest time without contact. Each duration carries a ± from the terrain's vertical error divided by the rate at which the body crosses the skyline. A crossing too shallow to time is reported as *grazing* instead of receiving a margin.

**Table 1:** The six mission sections and the checks behind them.

| Section | Question answered | Engine checked against |
|---|---|---|
| Ready | What is built, tested and delivered, and what is blocked? | State rules and the hash-chained audit log (tests) |
| Launch | When can the vehicle leave the pad for a landing the lighting allows? | Flown liftoffs of IM-1, IM-2 and Blue Ghost 1 |
| Launch weather | Which published weather rule would be broken, in how many model runs? | One year of station observations at the Cape |
| Trip | Where is the spacecraft, when is it in shadow, what radiation does it meet? | NASA reconstructions of four CLPS trips and LRO; NOAA's event list |
| Landing | How long are the Sun and Earth above the site's skyline? | Blue Ghost 1's reported end; golden fixtures |
| Contact | When can a ground station hear the lander, and how long must a message wait? | DSN masks and SPICE station kernels (tests) |
| Return | Where does the booster or a capsule come down, in what sea? | OSIRIS-REx's measured landing point |




**(a)** IM-2: the six sections above the open Landing section and its four numbers.




**(b)** Griffin-1's Launch section: the next liftoff carries the larger held-out miss, ±18 min.

**Figure 2:** Two screens at commit `9d1d5b1`, 1440 pixels wide, dark theme, cropped.

### 4.2 Launch: liftoff instants backed out from the landing

**Landing rule.** A landing time is allowed when the Sun's centre is above the site's skyline and rising, and the Earth is up. The project manager may narrow the Sun elevation to a band. Landing times that satisfy the rule are moved back by the time in lunar orbit and the flight time. The result is the set of days on which the vehicle must leave Earth.

**Liftoff instant.** For a direct transfer the arrival at the Moon fixes the plane of the parking orbit. The pad must lie in that plane at liftoff. The climb direction then follows from $\sin A = \cos i / \cos \varphi$. Here $A$ is the inertial azimuth, $i$ the parking orbit's inclination and $\varphi$ the pad's latitude. The angle a climb falls behind a circular orbit started at the pad, called here the *climb constant*, is measured on the vehicle's flown lunar launches. A design then corrects three quantities in the force model until the arrival matches. They are the liftoff time, the coast in the parking orbit before trans-lunar injection (TLI), and the speed that TLI adds. The force model holds the Sun, the Moon as a point mass and the Earth's zonal harmonics J2 to J6. A candidate climb azimuth outside the range's published or assumed limits is refused.

**Phasing loops.** Blue Ghost 1 flew several long Earth orbits before leaving for the Moon. For such a mission, the design raises the perigee at the first apogee. It then sets the period of the remaining loops and leaves Earth at the last perigee. It corrects liftoff, coast and the three burns in the force model for each launch day and keeps the cheapest loop count.

### 4.3 Trip: propagation, shadow and correction budget

A flown trip is read from NASA's reconstruction [2]. A designed trip is propagated in the J2000 frame on the barycentric dynamical time scale with an eighth-order Dormand–Prince integrator at a relative tolerance of $10^{-12}$. The Sun and Moon are point masses from DE440 [10]. Earth gravity is modelled with EGM2008 [32] to degree and order 8, capturing near-Earth tesseral harmonics. Within ten lunar radii (17,380 km) the Moon's field is the GRAIL model GRGM1200A [14] to degree 50. Solar radiation pressure is off by default, because no lander publishes its area and mass at that point in the flight. Shadow stretches come from conical Earth and Moon shadows with the overlap of the two discs [13], and each edge is bisected.

The correction budget is worked out for two sets of errors. The first is an assumed error model. In the second, the errors at departure come from the injection accuracy printed in the launch vehicle's user guide. The Falcon 9 guide prints rows for low Earth orbit and geosynchronous transfer orbit only, with none for trans-lunar injection, so its geosynchronous row serves as a proxy. Errors at departure, in orbit knowledge and in each burn (the model of Gates [33]) are pushed through the trajectory in two ways. The linear method uses state transition matrices of each stretch. The Monte Carlo method flies seeded runs through the full force model. In each run, every correction burn aims the run's own estimated state at the planned arrival. The answer reports the 95th and 99th percentiles of the total correction and labels them a budget for the errors named, which are assumed except where the launch vehicle's guide publishes them.

### 4.4 Trip: radiation screening

Belt fluence along the trajectory comes from AE8 and AP8 [21, 22] through IRBEM in SpacePy [23], with a factor of two on every value. IRBEM evaluates AP8 with the GSFC 12/66 magnetic field model at epoch 1970, preserving the original model calibration. The cosmic-ray dose is CRaTER's measured rate [25] scaled by the change of the model of Matthiä et al. [24] with solar modulation. The modulation parameter comes from the Oulu neutron monitor. On the surface, Chang'e-4's measured rate [26] is scaled the same way. It is also scaled by the fraction of open sky above the site, taken from the site's own skyline. Solar particle events during a flown trip are listed from NASA's DONKI record. Three past large events give comparison fluences: August 1972 from King's spectrum [27], and October 1989 and October 2003 integrated from

GOES proton data. Every radiation answer states that it is screening only and names the tools in which a design must be confirmed.

### 4.5 Contact: when a station hears the lander

A ground station hears the lander when two conditions hold at once. Earth must be above the lander's skyline, which is the surface engine's existing number. The lander must also be above the station's own horizon mask, taken from the DSN handbook [29] per degree of azimuth. The station positions come from SPICE station kernels. The union over the 14 catalogued DSN antennas is the *time a ground station can hear the lander*. Contacts between nodes, each with its one-way light time, feed an earliest-arrival search over contacts, the simplest exact form of contact graph routing [30]. It yields the longest time a message created on the lander waits before it reaches Earth.

The plan can be exported in the contact-plan format of the Interplanetary Overlay Network or as CSV. Planned relays (such as Lunar Pathfinder and Altus-1) are placed in the contact graph by sweeping node and phase over a $36 \times 36$ grid, reporting ranges of contact durations and store-and-forward wait times. Real lunar orbiters (LRO, Danuri, Chandrayaan-2, and CAPSTONE) are propagated from weekly JPL Horizons ephemerides as SPK type 13 kernels to display geometric overhead passes. For rovers, line-of-sight and first Fresnel zone clearance over the 20-meter LOLA terrain are computed up to 5 km, reporting clear, blocked, or uncertain states from elevation uncertainties. Link budgets remain uncomputed, because commercial transmitter powers and gains are not public.

### 4.6 Launch weather

The twelve weather rules of the Falcon 9 fact sheet [15] are stored verbatim as data, each with its unit, height and class. Two are *model* rules, checked against forecast numbers. Two are *proxy* rules, for which a model can only show the ingredients. Eight are *observation* rules, such as lightning within a set distance in the last 30 minutes, for which no forecast probability is given. A model rule is evaluated per run of the 31-member GEFS ensemble [16]. The wind at the rule's height of 162 ft is interpolated on a logarithmic profile between the model's 10 m and 80 m winds. The value is labelled as interpolated.

The result is a share, $k$ of 31 runs, with its exact interval. Single deterministic forecasts from the National Weather Service (NWS) and High-Resolution Rapid Refresh (HRRR) appear alongside the ensemble shares. A disagreement note flags when single models fall on opposite sides of the threshold or differ by more than HRRR's measured error. Beyond the ensemble's range, the answer falls back to month-by-hour climatology. Wind at 50 m comes from NASA POWER, which is based on MERRA-2 [19], and thunder hours come from GLM [18]. The range's official forecast is entered by the project manager with its document link and is never scraped.

### 4.7 Return: booster zone and capsule entry

The booster's landing type and zone come from the vehicle catalogue and Launch Library 2 landing records. For a planned droneship landing, the zone lies along the designed launch direction. The GFS-Wave forecast there is compared with limits that the team enters, since providers do not publish them. No CLPS mission returns a capsule. The capsule entry model is therefore checked on OSIRIS-REx [34]. It is a point mass with a ballistic coefficient. It falls from the entry interface through the NRLMSIS 2.1 empirical atmosphere [35], driven by the day's solar flux and geomagnetic indices, over a rotating Earth with J2 gravity. When space weather indices are unavailable, it falls back to the 1976 U.S. Standard Atmosphere [36]. It has no lift and no parachutes.

### 4.8 Ready: readiness, parts, risks and the audit log

Readiness is a state machine per item. Its states are not started, in build, in test, test passed, delivered to the lander provider, integrated, verified on the lander and ready. An item may also be blocked, with a reason, from any state. Every change needs a date and an evidence link, and illegal moves are refused. Parts carry their status and a dependency path to the measurements they serve. Risks use a configurable likelihood and consequence scale and are never multiplied or summed. Every write needs a named session and is recorded in an append-only log in which each entry holds the hash of the previous one. A database trigger rejects updates and deletions of that log. The landing history lists 14 lunar landing attempts as counts with sourced causes, never as a success probability.

### 4.9 Agents

Four language-model agents describe a site, rank sites against a stated requirement, write the same facts for three audiences and answer questions. They reach the system only through ten read-only tools, four of which read the mission sections. A section is cut to at most 200 numbers before an agent sees it, and every kept value keeps its margin. A guard checks every number and full date in an answer against that run's tool results. It also rejects verdict words such as *go*, *no-go* or *safe*. After one failed repair, the answer is replaced by a summary built only from measured values.

## 5 Evaluation

### 5.1 Setup and choice of references

The engine checks, response times, test counts and screens were measured with the codebase at commit `9d1d5b1`. The machine was an Apple M3 with eight cores and 8 GB of memory. The software was macOS 27.0, Python 3.12.14, PostgreSQL 16.15 and Redis 8.10.2. Every table below is printed by a script or a `make` target listed in appendix A. The references were chosen because they are the only public cases of each kind. NASA's reconstructions cover the four flown CLPS trips, and the Lunar Reconnaissance Orbiter (LRO) adds a well-tracked lunar orbit. OSIRIS-REx is the only recent capsule with both a published entry state and a measured landing point [37, 38]. The plan also named GMAT for the propagator and SPENVIS for the belts. An installation attempt of GMAT R2026a was conducted on this ARM64 macOS host. The distribution requires x86-64 binary translation, and its installer archive could not be retrieved within practical network limits. Closed-form analytic tests and NASA's own AE8/AP8 code serve as empirical replacements.

### 5.2 Landing: the Sun and Earth at the flown landers

Each flown lander's published LROC landing point was used as it stands, with its published surface dates. Nothing was tuned to these events. Blue Ghost 1 landed at 18.56° N, outside the polar terrain model, so its skyline is a smooth sphere. Its computed sunset falls on 16 March 2025 at 20:10 UTC, the date on which NASA reports that the mission ended. A golden test pins this sunset. IM-1 and IM-2 both ended while the Sun and Earth were still up at their sites. IM-1's next computed sunset is 2 March 2024 at 14:01 UTC ±28 min, with 83 % of the surrounding terrain inside the elevation model. IM-2's is 19 March 2025 at 19:17 UTC ±34 min. No cause is claimed for either end. The comparison shows that lighting geometry alone does not predict when a mission ends, and the product does not claim that it does.

At the pole the answer depends on metres of terrain. Table 2 gives the four numbers at three contrasting candidate sites over a reference window of 708.75 h from 14 March 2027. At Connecting Ridge the Sun's centre stays between −1.61° and −0.70° all month, yet the site is lit in 81 % of the samples (fig. 3). The site stands 1,957.5 m above the reference sphere, and its

skyline has a mean elevation of −2.37° and a highest point of −0.06°.

**Table 2:** The four numbers at three candidate sites, in hours, 14 March 2027 plus 708.75 h.

| Site | Sunlight | Contact with Earth | Longest night | Longest without contact |
|---|---|---|---|---|
| Connecting Ridge | $574.4 \pm 17.1$ | $430.3 \pm 5.3$ | $121.6 \pm 17.1$ | $278.5 \pm 5.3$ |
| Malapert massif | $394.2 \pm 7.3$ | $570.6 \pm 7.6$ | $\geq 287.8$ | $138.1 \pm 7.6$ |
| Faustini rim | $103.1 \pm 24.0$ | $278.6 \pm 33.9$ | 605.7, grazing | 430.2, grazing |

≥: a lower bound cut short by the window. Grazing: the crossing is too shallow to time, so no margin is given.

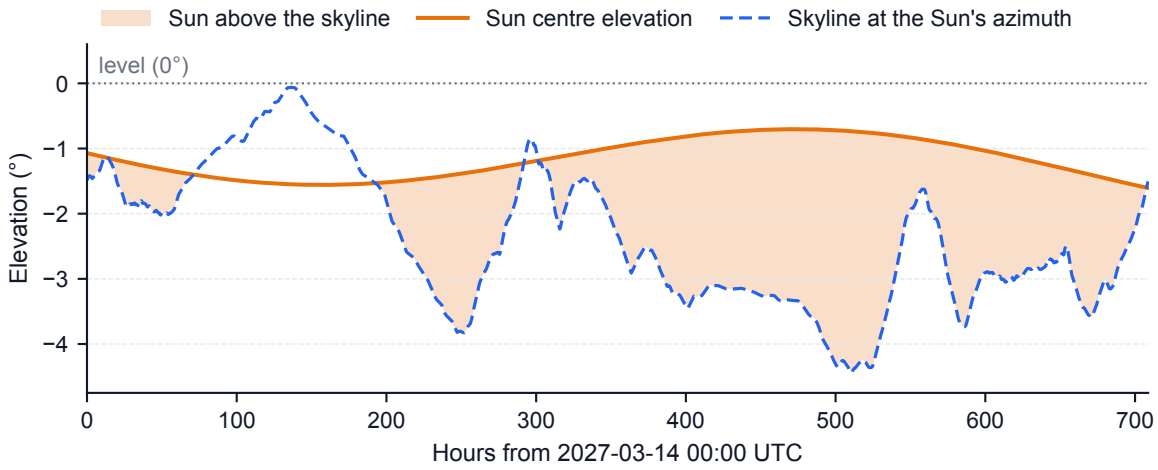



**Figure 3:** Sun and skyline at Connecting Ridge, 2,836 samples at 900 s. The Sun never rises above level but clears the lowered skyline in 81 % of samples.

### 5.3 Propagator and shadows against NASA's reconstructions

The acceleration implied by each reconstruction was compared every 10 minutes with the force model's. Where they differ by more than $1 \times 10^{-5}$ m/s$^2$, the spacecraft was firing thrusters. A *quiet stretch* is a run below that threshold, less 30 minutes at each end. From the start of a quiet stretch, the reconstructed state was propagated and compared with the reconstruction 24 or 72 hours later. Nothing was fitted. Table 3 gives the result.

**Table 3:** Position error after propagating a reconstructed state through a quiet stretch.

| Reference | Start (UTC) | Where | After | Error (km) | With assumed SRP (km) |
|---|---|---|---|---|---|
| Blue Ghost 1 | 2025-01-15 07:32 | cruise | 24 h | 0.37 | 0.51 |
| Blue Ghost 1 | 2025-01-16 07:32 | cruise | 24 h | 0.11 | 0.07 |
| Blue Ghost 1 | 2025-01-17 07:32 | cruise | 24 h | 0.11 | 0.07 |
| Blue Ghost 1 | 2025-01-15 07:32 | cruise | 72 h | 1.46 | 3.21 |
| Blue Ghost 1 | 2025-02-03 13:12 | cruise | 72 h | 1.08 | 0.62 |
| IM-1 | 2024-02-19 20:26 | cruise | 8.5 h | 0.64 | 0.64 |
| IM-2 | 2025-02-27 01:51 | cruise | 24 h | **4.40** | 4.22 |
| IM-2 | 2025-03-03 13:01 | lunar orbit | 24 h | 0.51 | 0.50 |
| IM-2 | 2025-03-04 13:01 | lunar orbit | 24 h | 0.27 | 0.28 |
| LRO | 2025-03-20 00:01 | lunar orbit | 24 h | 0.05 | 0.05 |
| LRO | 2025-03-21 01:41 | lunar orbit | 24 h | 0.03 | 0.03 |

SRP: solar radiation pressure with an assumed $C_r A/m$ of 0.01 m$^2$/kg. IM-1's longest quiet stretch is 8.5 h.

The plan's tolerances were 1 km after 24 h and 5 km after 72 h. Every row meets them except IM-2's cruise. The model alone gives 4.40 km, so the 1 km plan tolerance is not met. Modelling Earth's gravity with EGM2008 to degree and order 8 reduces this error from 4.62 km

by capturing near-Earth tesseral harmonics. Detailed analysis reveals that the remaining residual acceleration is piecewise-constant on a 30-minute barycentric dynamical time grid. It has a median magnitude of $6.71 \times 10^{-7}\,\mathrm{m/s^2}$ and points 58° away from the anti-solar direction. Two discrete velocity events appear in NASA's reconstruction at 19:00:00 UTC ($\Delta v = 0.060\,\mathrm{m/s}$) and 20:45:00 UTC ($\Delta v = 0.133\,\mathrm{m/s}$). Both events exhibit continuous positions and persistent one-second velocity jumps. They represent velocity changes of 0.060 m/s and 0.133 m/s (small burns or venting). If these two velocity changes were known in advance, applying them drops the 24-hour position error from 4.40 km to 1.65 km. Applying the piecewise-constant accelerations measured from the same reconstruction reduces the residual to 8.0 m. This calculation is not a model accuracy result. It demonstrates that the remaining 4.40 km is explained by those unmodelled accelerations.

In lunar orbit, perilune passes across 84 orbits were evaluated under different harmonic truncations of GRGM1200A. A point-mass lunar field leaves false velocity jumps of 0.5 to 1.6 m/s per pass, tripping the maneuver detection threshold of 0.05 m/s. Truncating the field at degree 50 suppresses the maximum pass residual to 16 mm/s (median below 10 mm/s) at a cost of 0.02 s per pass. Truncation at degree 100 lowers the residual to 0.5 to 3 mm/s but requires four to five times more runtime. Degree 50 is therefore the production setting. The harmonic switch-on radius is set to 10 lunar radii (17,380 km). This keeps eccentric orbits like Blue Ghost 1 (apoapsis at 4.28 lunar radii) entirely within the harmonic field and avoids 2 to 31 km of truncation boundary errors. A 24-hour forward and backward run returns to within 5.8 mm of its start.

Peregrine's reconstruction has no quiet stretch at all. Its unexplained acceleration has a median of $7.6 \times 10^{-5}\,\mathrm{m/s^2}$ and a 90th percentile of $1.8 \times 10^{-4}\,\mathrm{m/s^2}$. The provider traced the loss of that mission to a helium pressure control valve that failed to seal [39]. The residual shows where that leak appears in the trajectory; the engine does not estimate the leak.

Shadow edges were compared with SPICE's own occultation finder along two days of IM-2's lunar orbit, with the same spheres on both sides. Both found the same 24 shadow passes. The 48 edges differ by a median of 0.07 s and at most 0.10 s, inside the plan's 1 s.

### 5.4 Launch instants against the flown liftoffs

Each Falcon 9 lunar launch was designed from its pad, NASA's arrival at the Moon and the climb constant of the vehicle's other flights. Table 4 compares the designed liftoff with the published one under three choices of climb direction.

**Table 4:** Designed minus published liftoff, by the climb direction the design assumes.

| Mission | Flown climb azimuth | Due east | Azimuth it flew | Other flight's azimuth |
|---|---|---|---|---|
| IM-1 | 88.8° | 9.8 min | 0.2 min | −15.8 min |
| IM-2 | 86.6° | 27.4 min | −0.0 min | 17.5 min |
| Blue Ghost 1 | 110.2° | phasing loops (fig. 4) | | |

Designed at the azimuth each flight actually climbed, the liftoff lands within 0.2 min of the published one. The designed coast, TLI speed and flight time agree with the flown values within 1.4 min, 1 m/s and 0.02 d. That result uses the flight's own direction, which is not known before launch. Held out, each Nova-C lander designed at the other flight's azimuth misses by 15.8 min and 17.5 min. The plan asked for 10 min, so this criterion is **not met**. The two flown azimuths differ by 2.2°, and each degree moves the instant by about seven minutes. Nothing published predicts which azimuth a flight will take. The planned calendar therefore climbs at the mean azimuth of the vehicle's direct flights and states the larger held-out miss as its margin. For Griffin-1 the designed climb is 87.7°, and every liftoff carries ±18 min.

### 5.5 Phasing loops and the correction budget

Burns were read from Blue Ghost 1's reconstruction by propagating each half hour and comparing its end with the reconstruction. Three burns before departure were found: 22.53, 3.27 and 23.61 m/s, a total of 49.4 m/s. Replayed from the first reconstructed state with these burns, the six flown apogees and perigees fall within 2.9 min and 0.027 % in radius of the reconstruction. The plan asked for 1 h and 1 %.

The loop design was then run for eleven launch days around the flown one (fig. 4). The flown day needs three loops, as flown, and costs 50.3 m/s. It is the cheapest of the eleven days by 45 m/s. On that day the Sun's pull over each eight-day loop does most of the perigee raising, which a two-body design cannot capture. Designed at the azimuth Blue Ghost 1 actually climbed, 110.2°, the flown day's liftoff lands 0.6 min after the published one, inside the plan's 5 min.

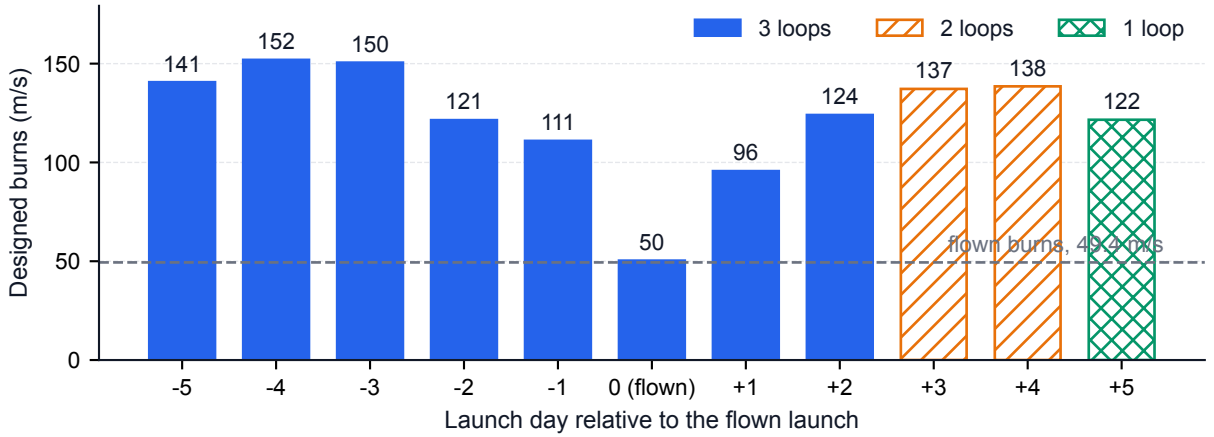



**Figure 4:** Designed burn total per launch day for Blue Ghost 1 (design climbing due east). The flown day is the cheapest; the dashed line is the total of the three flown burns.

Table 5 gives the correction budget for the three landers under assumed errors and under errors worked out from the published injection accuracy. That accuracy is the one the Falcon 9 user's guide prints for a geosynchronous transfer orbit (Rev 2, October 2015, Table 3-2, 3-sigma: apogee ±500 km, perigee ±10 km, inclination ±0.1°). Only that revision prints an injection accuracy, so it dates from the vehicle of 2015 and is not a figure for Block 5. The linear and Monte Carlo percentiles agree within 7.1 %, satisfying the 10 % acceptance limit. The published accuracy scales the 95th-percentile budget by a factor of 2.8 to 3.0 compared to the assumed errors. Blue Ghost 1's single flown correction of 1.15 m/s lies well inside its budget. IM-2's seven corrections total 24 m/s, falling well within its 95th-percentile envelope of 63.27 m/s. IM-1's reconstruction exhibits no mid-course corrections because its burns occurred during reconstruction data gaps.

**Table 5:** Total correction budget under assumed errors and published injection accuracy.

| Mission | Error model | 95th, lin / runs | 99th, lin / runs | 99th, 95 % interval | Arrival altitude |
|---|---|---|---|---|---|
| Blue Ghost 1 | Assumed | 14.64 / 14.16 | 19.24 / 18.08 | 16.52–19.85 | 156.2 to 160.2 km |
| | Published | 40.78 / 40.63 | 53.97 / 54.24 | 50.29–58.52 | 156.4 to 160.2 km |
| IM-1 | Assumed | 20.90 / 21.64 | 27.43 / 28.50 | 25.93–30.40 | 110.6 to 114.5 km |
| | Published | 61.96 / 60.13 | 81.47 / 77.24 | 70.85–81.43 | 110.6 to 114.6 km |
| IM-2 | Assumed | 21.29 / 21.45 | 27.84 / 28.89 | 26.24–30.24 | 99.8 to 103.8 km |
| | Published | 62.71 / 63.27 | 82.40 / 88.66 | 80.00–94.73 | 99.7 to 103.9 km |

Velocities in m/s. 1,000 seeded Monte Carlo runs compared against 100,000-draw linear covariance propagation.

### 5.6 Radiation screening

SPENVIS was not run. The engine's belt fluences were therefore compared with a second implementation of the same models, NASA SPDF's own AE8/AP8 code. Each code uses its own magnetic field shells. The comparison used one-minute samples within 15 Earth radii along each flown trajectory, with solar-maximum maps (table 6). Electrons agree within 5 % on every trip. Protons agree within 5 % on the two trips that spend hours in the inner belt. IM-2's proton fluence, about a thousand times smaller, differs by 40.5 %. This discrepancy stems from the magnetic field model epoch rather than the transport algorithms. IRBEM evaluates AP8 using the GSFC 12/66 field model at epoch 1970, preserving the calibration of the original maps. The SPDF driver uses the International Geomagnetic Reference Field evaluated at the 2025 flight epoch. When both codes are supplied with identical magnetic coordinates, their computed fluences agree to 0.0 % on every trajectory. IM-1's reconstruction starts 242,208 km from Earth, beyond the belts, so its belt numbers are reported as missing with that explicit reason.

**Table 6:** Belt fluence along flown trips: the engine (IRBEM) against NASA SPDF's AE8/AP8.

| | | Electrons >1 MeV ($1/\mathrm{cm}^2$) | | | Protons >10 MeV ($1/\mathrm{cm}^2$) | | |
|---|---|---|---|---|---|---|---|
| Mission | Samples | IRBEM | SPDF | Diff. | IRBEM | SPDF | Diff. |
| IM-2 | 437 | $1.09 \times 10^{10}$ | $1.13 \times 10^{10}$ | −2.8 % | $8.55 \times 10^{5}$ | $1.44 \times 10^{6}$ | −40.5 % |
| Blue Ghost 1 | 3,311 | $2.04 \times 10^{10}$ | $2.15 \times 10^{10}$ | −5.0 % | $8.00 \times 10^{8}$ | $7.94 \times 10^{8}$ | 0.7 % |
| Peregrine | 867 | $2.35 \times 10^{10}$ | $2.33 \times 10^{10}$ | 0.8 % | $2.67 \times 10^{8}$ | $2.80 \times 10^{8}$ | −4.8 % |

The cosmic-ray model was checked in absolute terms for June 2009. Its dose in a thin silicon layer in free space is 14.1 cGy per year. Two of CRaTER's detectors gave 18.9 and 16.5 cGy per year, so the model is lower by 26 % and 15 % [25]. Both lie inside the stated ±30 %. Scaled to each trip, the unshielded cosmic-ray dose is $(0.21 \pm 0.06)$ mGy per day for IM-1 and $(0.17 \pm 0.05)$ mGy per day for Blue Ghost 1. The GOES peaks of the October 1989 and October 2003 events match NOAA's list within 5.5 % and 0.0 %, inside the plan's 10 %.

### 5.7 Capsule entry against OSIRIS-REx

The OSIRIS-REx capsule's reconstructed trajectory in NAIF's archive ends at the entry interface, 6,503.142 km from the Earth's centre, at 14:41:55 UTC on 24 September 2023. Its state there matches the navigation team's published entry state to the last printed digit: flight path angle −8.184°, heading 66.396° and Earth-relative speed 12.361 km/s [37]. The ballistic coefficient $\beta$ was built from three published values. The first two are the capsule's mass at entry and the diameter of the Stardust forebody it shares. The third is Stardust's hypersonic axial force coefficient $C_A$ of 1.48 to 1.56 [40]. Atmospheric entry was simulated using NRLMSIS 2.1 [35], parameterized by the day's solar radio flux ($F_{10.7} = 174.1$ sfu) and geomagnetic index ($a_p = 4.0$) from the OMNI2 archive. Table 7 compares the modelled landing with the point measured by the recovery team [38].

NRLMSIS 2.1 was chosen over NRLMSISE-00 [41] and the 1976 U.S. Standard Atmosphere by comparison with the air measured at 30 to 70 km on 24 September 2023. The measurements are of two kinds. The first is the limb profiles of the TIMED satellite's SABER radiometer [42]. The second is the profiles of the Table Mountain Facility lidar for nine nights from 19 to 30 September. The lidar profiles come from the archive of the Network for the Detection of Atmospheric Composition Change (NDACC) [43]. Density decides the choice, not temperature, because density sets how the capsule slows. Between 30 and 70 km, the density of NRLMSIS 2.1 differs from the SABER profiles by 0.6 % and from the lidar profiles by 1.6 %. For NRLMSISE-00 the figures are 3.7 % and 5.0 %. All four are root mean square values over eight 5 km bands. By band, NRLMSISE-00 is warmer than the measurements by 0.6 to 9.5 K (root mean square

4.4 and 5.1 K) and NRLMSIS 2.1 by 0.2 to 5.3 K. NRLMSIS 2.1 has the lower density difference in every cut of the profiles: by latitude, by local solar time and with each lidar night left out.

These measurements are only partly independent of the models, and four cautions apply. First, the selection rule was written after both landing results were known. Second, NRLMSIS 2.0, on which NRLMSIS 2.1 builds, was fitted to SABER temperatures from 2002 to 2016 [44]. Third, the lidar density is scaled to an ancillary profile that its file names as MERRA-2 [19]. MERRA-2 is in the NRLMSIS 2.0 fit up to 55 km. The level of the lidar density is therefore not independent of that fit, but its change with height is. Fourth, the producer marks the SABER files “Unvalidated”.

**Table 7:** Modelled OSIRIS-REx landing against the measured point (limit 10 km).

| $C_A$ | $\beta$ (kg/m$^2$) | Landing | Miss (km) | To ground (s) | Peak (g) |
|---|---|---|---|---|---|
| 1.48 | 64.8 | 40.3687° N, 113.2861° W | 3.9 | 525 | 31.9 |
| 1.52 | 63.1 | 40.3612° N, 113.3169° W | **6.6** | 531 | 32.0 |
| 1.56 | 61.5 | 40.3539° N, 113.3468° W | 9.3 | 536 | 32.2 |

Peak: the largest deceleration on the way down. Modelled with NRLMSIS 2.1 driven by daily space weather indices.

At the nominal drag coefficient ($C_A = 1.52$), the landing point misses by 6.6 km. Over the published $C_A$ range, the miss spans 3.9 to 9.3 km. The plan’s acceptance limit of 10 km is therefore **met** across the entire drag range. Under the simplified 1976 U.S. Standard Atmosphere [36], the nominal miss is 9.5 km (6.7 to 12.2 km). Under older NRLMSISE-00, the miss increases to 12.1 km (9.4 to 14.8 km), failing the 10-kilometre limit. The mission team’s pre-flight simulation from the identical entry state missed by 8.9 km [38]. Their day-of-entry reconstruction using local weather balloons reached 3.5 km. A seeded Monte Carlo of 200 runs with NRLMSIS 2.1 yields a 99 percent dispersion ellipse of 51.1 km by 21.5 km that holds the measured landing point.

### 5.8 Launch weather against station observations

No public meteorological tower reports winds at the rule’s 162 ft level. Verification was therefore performed at 10 m elevation at Cape Canaveral Space Force Station (KXMR) and Vandenberg Space Force Base (KVBG). The rule’s limit of 30 mph at 49 m translates to 23.5 mph (10.52 m/s) at 10 m using a neutral logarithmic wind profile.

Multi-year verification was conducted using the NOAA GEFSv12 reforecast archive over 2000 to 2019 (1,583 days evaluated). At Vandenberg, the ensemble demonstrates positive skill: with 75 threshold exceedance events at 24 h lead time, the Brier skill score is 0.30 (95 % confidence interval 0.16 to 0.41), and the continuous ranked probability score (CRPS) skill is 0.33 (table 8). At 48 h lead time, with 74 events, the Brier skill is 0.24 (0.09 to 0.38). However, Vandenberg does not pass the acceptance gate on reliability. The bin for a forecast share of 1.0 (all five runs over the limit) holds 41 cases across both leads. The wind was above the limit in 36 of them (88 %, 95 % confidence interval 75 to 95 %). This discrepancy of 0.12 against the forecast share exceeds the 0.10 tolerance allowed. At Cape Canaveral, only one event occurred at 24 h and three at 48 h across 1,459 evaluated runs. This is fewer than the 10 events required for a reliable Brier skill calculation. The runs’ wind is worse than the station’s own record: the CRPS skill is negative ($-0.97$), and the runs average 5.28 m/s against the station’s 3.11 m/s. The likely reason is that the 0.25-degree cell at the Cape is partly sea, where the wind is stronger than at a station inland. Because neither pad passes the acceptance gate, launch weather probability shares remain marked as not checked against observations at both sites.

Operational single-model forecasts from the High-Resolution Rapid Refresh (HRRR) were evaluated across 1,826 runs from 2021 to 2025 at leads of 1, 6, 12, and 18 h. HRRR achieves a mean absolute error of 1.0 to 1.4 m/s and a bias of 0.0 to 0.6 m/s. This is 15 to 42 % closer to observed winds than climatology. At 1 h lead at Vandenberg, HRRR achieves a CRPS skill

**Table 8:** GEFSv12 reforecast 10-meter wind verification against station observations.

| Station | Lead | Cases | Events | Brier skill (95 % CI) | CRPS skill | Passed gate |
|---|---|---|---|---|---|---|
| Vandenberg (KVBG) | 24 h | 1,080 | 75 | 0.30 (0.16 to 0.41) | 0.33 | no (reliability) |
| Vandenberg (KVBG) | 48 h | 1,082 | 74 | 0.24 (0.09 to 0.38) | 0.33 | no (reliability) |
| Cape Canaveral (KXMR) | 24 h | 1,459 | 1 | insufficient events | −0.97 | no (bias) |
| Cape Canaveral (KXMR) | 48 h | 1,482 | 3 | insufficient events | −0.99 | no (bias) |

of 0.19 (0.14 to 0.22) and a Brier skill of 0.16 (0.01 to 0.32). Over all four leads combined, the over/under Brier skill is 0.06 at Vandenberg and −0.32 at Cape Canaveral.

### 5.9 Lunar orbiters and rover line of sight

Geometric passes for real lunar orbiters (LRO, Danuri, Chandrayaan-2, and CAPSTONE) were validated against JPL Horizons observer ephemerides across landing sites at IM-2, IM-1, and Blue Ghost 1 (table 9). Azimuth and elevation agree within 0.0010° (plan tolerance 0.02°). Rise and set times over a spherical horizon agree within 0.003 s (plan tolerance 2.0 s). Checking LRO line-of-sight vectors against NAIF's definitive reconstruction kernel (`lrorg_2025074_2025166`) gives a maximum angular separation of 0.00018° while above the horizon (plan tolerance 0.01°).

**Table 9:** Lunar orbiter visibility checks against JPL Horizons and NAIF reconstruction.

| Site | Orbiter | Up (min) | Max angle | Max Δel | Max edge $\Delta t$ |
|---|---|---|---|---|---|
| IM-2 | LRO | 82 | 0.0002° | 0.0002° | 1.6 ms |
| IM-2 | Danuri | 57 | 0.0004° | 0.0003° | 1.8 ms |
| IM-2 | Chandrayaan-2 | 84 | 0.0002° | 0.0002° | 2.0 ms |
| IM-2 | CAPSTONE | 766 | <0.0001° | <0.0001° | no edges |
| Blue Ghost 1 | LRO | 64 | 0.0010° | 0.0006° | 3.1 ms |
| IM-2 | NAIF LRO kernel | 71 | 0.00018° | 0.00017° | not applicable |

Rover-to-lander line of sight was evaluated on the 20-meter LOLA digital elevation model for IM-2 (Yaoki rover) and Griffin-1 (FLIP rover). Ray traces were sampled every 5 degrees in azimuth out to radii of 500 and 2,000 m (1,440 cells per test). The running maximum elevation algorithm was compared with an independent brute-force line test sampled every meter. Outside the elevation model's vertical uncertainty band, agreement is 100.00 % at both sites (plan tolerance 99 %). Within 500 m, the 2 m vertical DEM error creates an angular uncertainty wider than the antenna elevation angle, causing 1,219 of 1,440 cells to be reported as uncertain. The planner correctly serves this as a wide probability range rather than an unsubstantiated certainty.

### 5.10 Response times and the compute queue

Table 10 gives response times through the API at commit `9d1d5b1`, with the compute and feeds workers stopped. The run followed three readings of the one-minute load average below 2, taken 20 s apart. A *cold* surface request asks for a window that no earlier request has asked for. It counts only if the response reports a cache miss, and all 30 of each kind did. A *warm* request repeats a window or reads a stored section answer. Each section was read 20 times for each of four missions: IM-1, IM-2, Blue Ghost 1 and Griffin-1.

A stored answer of any section returns within 10.4 ms at the 95th percentile. Of the 11.2 ms of a cold four-numbers request, the positions of the Sun and Earth take 6.7 ms and the two tracks against the skyline 2.6 ms. Most of a cold request is therefore spent on the ephemeris, not on the terrain.

A trip, launch calendar or contact answer that is not yet stored is computed as a job (section 3). The load test asks for these three sections of all 20 missions in one burst. Extra

**Table 10:** Response times through the API, median and 95th percentile.

| Request | *N* | Median | p95 |
|---|---|---|---|
| *Landing: the surface engine* | | | |
| Horizon mask, 1,440 bearings (precompute) | 75 | 45.9 ms | 50.6 ms |
| Four numbers, cold / warm | 30 / 30 | 11.2 / 0.83 ms | 14.4 / 0.92 ms |
| Timeline, 210 kB, cold / warm | 30 / 30 | 16.6 / 4.05 ms | 19.3 / 4.38 ms |
| *Stored section answers, warm* | | | |
| Six-section summary | 80 | 1.3 ms | 1.4 ms |
| Ready | 80 | 1.5 ms | 1.9 ms |
| Launch | 80 | 3.0 ms | 8.1 ms |
| Launch weather | 80 | 5.4 ms | 10.4 ms |
| Trip | 80 | 9.3 ms | 9.9 ms |
| Contact | 80 | 2.0 ms | 2.3 ms |
| Return | 80 | 2.4 ms | 4.4 ms |

launch and contact windows bring the burst to 138 requests. It runs against its own copy of the database, with two compute workers on the eight-core machine and the backend at `f421298`. The API answered 43 requests from stored answers and queued 58 jobs. It refused 37 contact requests with status 409 and a reason. The 13 missions concerned have no landing to hear. Their landing spot is not published, the lander was lost before reaching the Moon, or the delivery was cancelled. With all 138 requests in flight at once, the API's median response was 761 ms and its slowest 1.42 s. The two workers emptied the queue in 211.5 s, 16.5 jobs per minute, and were busy for 89 % of that time. No job failed or was retried. At the median a contact job ran for 0.19 s, a launch calendar for 13.3 s and a trip for 55.0 s. The longest wait in the queue was a trip's, 160.7 s. The same burst repeated afterwards found every answer stored. It received 101 answers and the same 37 refusals, at a median of 318 ms per request.

### 5.11 Tests and accessibility

On the backend at commit `9d1d5b1`, the code linter, strict static type checking of 281 source files and all ten architectural import contracts pass without warnings. The full test suite collects 836 tests. A total of 619 fast unit tests run without external service dependencies in 18 s. Another 181 integration tests execute against PostgreSQL. The 42 golden tests among them pin model behaviors to committed fixtures and flown trajectories.

On the frontend, static type checks pass, and 281 unit and component tests pass across 24 test files in Vitest. All 60 browser end-to-end tests pass in Chromium in 4.9 min against the running API. Twenty-four of them validate against recorded contract fixtures, and 13 test live API responses. Four automated browser tests execute the axe accessibility engine against levels A and AA of the Web Content Accessibility Guidelines (WCAG) 2.1. Across 26 desktop routes, 14 expanded view pages, the settings dialog, and 14 mobile viewport routes, zero serious or critical accessibility violations were detected.

## 6 Limitations

**Acceptance tolerances not met.** Table 11 lists planned acceptance tolerances alongside measured results. Four items fall outside the plan. First, designing a launch instant with an assumed climb azimuth from another flight misses by 15.8 min and 17.5 min against the planned 10 min tolerance, because launch vehicle ascent profiles are proprietary. Second, the model alone misses IM-2's reconstructed cruise by 4.40 km after 24 hours, against the planned 1 km. The miss would be 1.65 km if the two velocity changes of 0.060 m/s and 0.133 m/s (small burns or venting) had been known in advance. Third, the weather ensemble fails the acceptance gate at both launch ranges. Cape Canaveral exhibits negative CRPS skill (−0.97) and has insufficient

events for Brier skill. Vandenberg achieves the required Brier skill (0.30 and 0.24) but fails the reliability threshold (0.12 discrepancy against the 0.10 limit in the share-of-1.0 bin). Fourth, IM-2 belt protons differ by 40.5 % between IRBEM and SPDF due to magnetic field model epochs (agreeing to 0.0 % on identical coordinates). Capsule entry with NRLMSIS 2.1 meets the 10 km plan limit (landing within 6.6 km, range 3.9 to 9.3 km), but a rule written after both landing results were known chose that air model. NRLMSISE-00 missed by 12.1 km, with a density above the measured density at 30 to 70 km.

**Table 11:** Planned acceptance tolerances and the measured values.

| Check | Plan | Measured | Met |
|---|---|---|---|
| Propagation of reconstructions, 24 h | 1 km | 0.11–0.51 km; IM-2 cruise 4.40 km | partly |
| Propagation of reconstructions, 72 h | 5 km | 1.08, 1.46 km | yes |
| Lunar orbit gravity field per pass | 0.05 m/s | 16 mm/s (degree 50); degree 0 leaves 0.5–1.6 m/s | yes |
| LRO in lunar orbit, 24 h | 1 km | 0.03, 0.05 km | yes |
| Forward and backward run, 24 h | 1 m | 5.8 mm | yes |
| Shadow edges against SPICE | 1 s | 0.10 s | yes |
| Liftoff from the flown TLI | 2 or 5 min | 4 to 15 s | yes |
| Liftoff designed at the flown azimuth | 2 or 5 min | 0.0 to 0.6 min | yes |
| Liftoff designed at the other flight's azimuth | 10 min | 15.8, 17.5 min | no |
| Loop replay: apsis time, radius | 1 h, 1 % | 2.9 min, 0.027 % | yes |
| Loop design: flown loop count and date | both | 3 loops; period holds the date | yes |
| Correction budget, linear against runs | 10 % | 7.1 % (published injection accuracy) | yes |
| Belt fluence, two AE8/AP8 codes | 5 % | electrons 5.0 %; IM-2 protons 40.5 % | partly |
| Cosmic-ray dose against CRaTER | 30 % | −26 %, −15 % | yes |
| Solar event peaks against NOAA | 10 % | 5.5 %, 0.0 % | yes |
| Wind shares, multi-year gate | Brier $\geq 0.1$, rel $\leq 0.1$ | KVBG fails rel (0.12); KXMR negative CRPS (−0.97) | no |
| Capsule landing | 10 km | 6.6 km (NRLMSIS 2.1, range 3.9–9.3 km; NRLMSISE-00 12.1 km) | yes |
| Lunar orbiters against Horizons | 0.02°, 2.0 s | 0.0010°, 0.003 s (NAIF check 0.00018°) | yes |
| Rover line of sight | 99 % | 100.00 % outside uncertain band | yes |
| GMAT and SPENVIS comparisons | run | not run | no |

**Parts of the plan not built.** Commercial lander operators do not publish transmitter power, receiver sensitivity, or antenna gains. Communications modeling therefore evaluates geometric line of sight rather than RF link budgets. No CLPS mission currently returns payloads from the lunar surface to Earth, and planned missions publish no Earth-return trajectory or atmospheric entry state. The Moon-to-Earth return trajectory and capsule recovery site weather are therefore not built. European meteorological ensemble forecasts (ECMWF) are not archived because open data access provides only global grids rather than point extractions. NASA GEOS-FP forecasts and NAVAREA IV maritime hazard broadcasts could not be accessed from the automated feeds worker.

**Validity of the references.** GMAT and SPENVIS, the plan's references, were not run. Closed-form tests replace GMAT: two-body motion against Kepler's equation, the J2 nodal drift against its analytic rate, and forward-and-backward runs. NASA's AE8/AP8 code replaces SPENVIS, but it implements the same models, so agreement shows correct use of the models, not their accuracy. The four CLPS reconstructions are orbit determination products, not truth, and IM-2's contains an unexplained acceleration.

**Carried over from the surface engine.** The terrain's vertical error is one value per product, 2 m or 4 m, not a per-pixel error, and crossing errors are treated as independent. The near-field noise threshold is not part of any margin. Terrain is modelled only south of 80° S; elsewhere the skyline is a smooth sphere and the durations are upper bounds. No model key was configured on the measurement machine, so the agents were not evaluated with a live language model. Their tests use scripted model replies. No user with a screen reader has tested the screens.

## 7 Conclusion

POLAR WINDOW follows a CLPS delivery across six operational sections, from ground readiness to post-flight recovery. Every computed number provides provenance, freshness, and explicit uncertain bounds or missing reasons. The system produces no arbitrary go/no-go verdicts or composite scores. Physical models were validated against the public record of flown missions, including NASA reconstructions of four CLPS trajectories, published liftoffs, and satellite observations. Most acceptance checks satisfy the planned tolerances. Four measured results fall outside the plan: the liftoff designed at another flight's azimuth, the IM-2 cruise position error, the wind check at both launch pads and the IM-2 belt protons. The GMAT and SPENVIS comparisons were not run. The capsule entry model meets its tolerance. Planned relays, lunar orbiters, and rover line of sight provide transparent communication visibility. RF link budgets and the Moon-to-Earth return trajectory remain unbuilt because required engineering parameters are not public. The system serves as a screening and comparison tool, designed to help mission teams evaluate trade-offs before committing to specialized analysis pipelines.

## References

[1] NASA Science Mission Directorate. CLPS deliveries. NASA Science. URL `https://science.nasa.gov/lunar-science/clps-deliveries/`. Accessed 2026-09-27.

[2] NASA Navigation and Ancillary Information Facility. CLPS SPICE archive (PDS4). Planetary Data System. URL `https://naif.jpl.nasa.gov/pub/naif/pds/pds4/clps/clps_spice/`. Reconstructed trajectories of Peregrine, IM-1, Blue Ghost Mission 1 and IM-2; accessed 2026-09-26.

[3] NASA Space Apps Challenge. CLPS lunar mission browser. Challenge statement, 2026. URL `https://www.spaceappschallenge.org/2026/challenges/clps-lunar-mission-browser/`.

[4] David E. Smith, Maria T. Zuber, Glenn B. Jackson, John F. Cavanaugh, Gregory A. Neumann, Haris Riris, Xiaoli Sun, Ronald S. Zellar, Craig Coltharp, Joseph Connelly, Richard B. Katz, Igor Kleyner, Peter Liiva, Adam Matuszeski, Erwan M. Mazarico, Jan F. McGarry, Anne-Marie Novo-Gradac, Melanie N. Ott, Carlton Peters, Luis A. Ramos-Izquierdo, Lawrence Ramsey, David D. Rowlands, Stephen Schmidt, V. Stanley Scott, George B. Shaw, James C. Smith, Joseph-Paul Swinski, Mark H. Torrence, Glenn Unger, Anthony W. Yu, and Thomas W. Zagwodzki. The lunar orbiter laser altimeter investigation on the lunar reconnaissance orbiter mission. *Space Science Reviews*, 150(1–4):209–241, 2010. doi: 10.1007/s11214-009-9512-y.

[5] E. Mazarico, G.A. Neumann, D.E. Smith, M.T. Zuber, and M.H. Torrence. Illumination conditions of the lunar polar regions using lola topography. *Icarus*, 211(2):1066–1081, 2011. doi: 10.1016/j.icarus.2010.10.030.

[6] Emerson J. Speyerer and Mark S. Robinson. Persistently illuminated regions at the lunar poles: Ideal sites for future exploration. *Icarus*, 222(1):122–136, 2013. doi: 10.1016/j.icarus.2012.10.010.

[7] P. Gläser, F. Scholten, D. De Rosa, R. Marco Figuera, J. Oberst, E. Mazarico, G.A. Neumann, and M.S. Robinson. Illumination conditions at the lunar south pole using high resolution digital terrain models from lola. *Icarus*, 243:78–90, 2014. doi: 10.1016/j.icarus.2014.08.013.

[8] Charles H. Acton. Ancillary data services of nasa's navigation and ancillary information facility. *Planetary and Space Science*, 44(1):65–70, 1996. doi: 10.1016/0032-0633(95)00107-7.

[9] Charles Acton, Nathaniel Bachman, Boris Semenov, and Edward Wright. A look towards the future in the handling of space science mission geometry. *Planetary and Space Science*, 150:9–12, 2018. doi: 10.1016/j.pss.2017.02.013.

[10] Ryan S. Park, William M. Folkner, James G. Williams, and Dale H. Boggs. The jpl planetary and lunar ephemerides de440 and de441. *The Astronomical Journal*, 161(3):105, 2021. doi: 10.3847/1538-3881/abd414.

[11] Steven P. Hughes, Rizwan H. Qureshi, Steven D. Cooley, and Joel J. Parker. Verification and validation of the general mission analysis tool (gmat). In *AIAA/AAS Astrodynamics Specialist Conference*. American Institute of Aeronautics and Astronautics, 2014. doi: 10.2514/6.2014-4151.

[12] Dario Izzo. Revisiting lambert's problem. *Celestial Mechanics and Dynamical Astronomy*, 121(1):1–15, 2015. doi: 10.1007/s10569-014-9587-y.

[13] Oliver Montenbruck and Eberhard Gill. *Satellite Orbits*. Springer Berlin Heidelberg, Berlin, Heidelberg, 2000. doi: 10.1007/978-3-642-58351-3.

[14] Maria T. Zuber, David E. Smith, David H. Lehman, Tom L. Hoffman, Sami W. Asmar, and Michael M. Watkins. Gravity recovery and interior laboratory (grail): Mapping the lunar interior from crust to core. *Space Science Reviews*, 178(1):3–24, 2013. doi: 10.1007/s11214-012-9952-7.

[15] NASA Kennedy Space Center. Falcon 9 crew dragon launch weather criteria. Technical Report FS-2020-05-568-KSC, National Aeronautics and Space Administration, 2020. URL `https://www.nasa.gov/wp-content/uploads/2021/07/falcon9_crewdragon_launch_weather_criteria_fact_sheet.pdf`.

[16] Hong Guan, Yuejian Zhu, Eric Sinsky, Bing Fu, Wei Li, Xiaqiong Zhou, Xianwu Xue, Dingchen Hou, Jiayi Peng, M. M. Nageswararao, Vijay Tallapragada, Thomas M. Hamill, Jeffrey S. Whitaker, Gary Bates, Philip Pegion, Sherrie Frederick, Matthew Rosencrans, and Arun Kumar. Gefsv12 reforecast dataset for supporting subseasonal and hydrometeorological applications. *Monthly Weather Review*, 150(3):647–665, 2022. doi: 10.1175/mwr-d-21-0245.1.

[17] Steven J. Goodman, Richard J. Blakeslee, William J. Koshak, Douglas Mach, Jeffrey Bailey, Dennis Buechler, Larry Carey, Chris Schultz, Monte Bateman, Eugene McCaul, and Geoffrey Stano. The goes-r geostationary lightning mapper (glm). *Atmospheric Research*, 125–126:34–49, 2013. doi: 10.1016/j.atmosres.2013.01.006.

[18] Katrina Virts, Steven J. Goodman, and Scott Rudlosky. Geostationary lightning mapper (GLM) combined hourly thunder hour. NASA Global Hydrometeorology Resource Center DAAC, 2025.

[19] Ronald Gelaro, Will McCarty, Max J. Suárez, Ricardo Todling, Andrea Molod, Lawrence Takacs, Cynthia A. Randles, Anton Darmenov, Michael G. Bosilovich, Rolf Reichle, Krzysztof Wargan, Lawrence Coy, Richard Cullather, Clara Draper, Santha Akella, Virginie Buchard, Austin Conaty, Arlindo M. da Silva, Wei Gu, Gi-Kong Kim, Randal Koster, Robert Lucchesi, Dagmar Merkova, Jon Eric Nielsen, Gary Partyka, Steven Pawson, William Putman, Michele Rienecker, Siegfried D. Schubert, Meta Sienkiewicz, and Bin Zhao. The modern-era retrospective analysis for research and applications, version 2 (merra-2). *Journal of Climate*, 30(14):5419–5454, 2017. doi: 10.1175/jcli-d-16-0758.1.

[20] GLENN W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078<0001:vofeit>2.0.co;2.

[21] James I. Vette. The AE-8 trapped electron model environment. Technical Report NSSDC/WDC-A-R&S 91-24 (NASA-TM-107820), National Space Science Data Center, NASA Goddard Space Flight Center, Greenbelt, MD, 1991. URL `https://ntrs.nasa.gov/citations/19920014985`.

[22] Donald M. Sawyer and James I. Vette. AP-8 trapped proton environment for solar maximum and solar minimum. Technical Report NSSDC/WDC-A-R&S 76-06 (NASA-TM-X-72605), National Space Science Data Center, NASA Goddard Space Flight Center, Greenbelt, MD, 1976. URL `https://ntrs.nasa.gov/citations/19770012039`.

[23] Jonathan T. Niehof, Steven K. Morley, Daniel T. Welling, and Brian A. Larsen. The spacepy space science package at 12 years. *Frontiers in Astronomy and Space Sciences*, 9:1023612, 2022. doi: 10.3389/fspas.2022.1023612.

[24] Daniel Matthiä, Thomas Berger, Alankrita I. Mrigakshi, and Günther Reitz. A ready-to-use galactic cosmic ray model. *Advances in Space Research*, 51(3):329–338, 2013. doi: 10.1016/j.asr.2012.09.022.

[25] N. A. Schwadron, T. Baker, B. Blake, A. W. Case, J. F. Cooper, M. Golightly, A. Jordan, C. Joyce, J. Kasper, K. Kozarev, J. Mislinski, J. Mazur, A. Posner, O. Rother, S. Smith, H. E. Spence, L. W. Townsend, J. Wilson, and C. Zeitlin. Lunar radiation environment and space weathering from the cosmic ray telescope for the effects of radiation (crater). *Journal of Geophysical Research: Planets*, 117(E12):E00H13, 2012. doi: 10.1029/2011je003978.

[26] Shenyi Zhang, Robert F. Wimmer-Schweingruber, Jia Yu, Chi Wang, Qiang Fu, Yongliao Zou, Yueqiang Sun, Chunqin Wang, Donghui Hou, Stephan I. Böttcher, Sönke Burmeister, Lars Seimetz, Björn Schuster, Violetta Knierim, Guohong Shen, Bin Yuan, Henning Lohf, Jingnan Guo, Zigong Xu, Johan L. Freiherr von Forstner, Shrinivasrao R. Kulkarni, Haitao Xu, Changbin Xue, Jun Li, Zhe Zhang, He Zhang, Thomas Berger, Daniel Matthiä, Christine E. Hellweg, Xufeng Hou, Jinbin Cao, Zhen Chang, Binquan Zhang, Yuesong Chen, Hao Geng, and Zida Quan. First measurements of the radiation dose on the lunar surface. *Science Advances*, 6(39):eaaz1334, 2020. doi: 10.1126/sciadv.aaz1334.

[27] JOSEPH H. King. Solar proton fluences for 1977-1983 space missions. *Journal of Spacecraft and Rockets*, 11(6):401–408, 1974. doi: 10.2514/3.62088.

[28] Robert C. Singleterry, Steve R. Blattnig, Martha S. Clowdsley, Garry D. Qualls, Chris A. Sandridge, Lisa C. Simonsen, Tony C. Slaba, Steven A. Walker, Francis F. Badavi, Jan L. Spangler, Aric R. Aumann, E. Neal Zapp, Robert D. Rutledge, Kerry T. Lee, Ryan B.

Norman, and John W. Norbury. Oltaris: On-line tool for the assessment of radiation in space. *Acta Astronautica*, 68(7–8):1086–1097, 2011. doi: 10.1016/j.actaastro.2010.09.022.

[29] Jet Propulsion Laboratory. DSN telecommunications link design handbook. Technical Report 810-005, Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA. URL `https://deepspace.jpl.nasa.gov/dsndocs/810-005/`. Module 301, antenna horizon masks; accessed 2026-09-27.

[30] Giuseppe Araniti, Nikolaos Bezirgiannidis, Edward Birrane, Igor Bisio, Scott Burleigh, Carlo Caini, Marius Feldmann, Mario Marchese, John Segui, and Kiyohisa Suzuki. Contact graph routing in dtn space networks: overview, enhancements and performance. *IEEE Communications Magazine*, 53(3):38–46, 2015. doi: 10.1109/mcom.2015.7060480.

[31] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.

[32] Nikolaos K. Pavlis, Simon A. Holmes, Steve C. Kenyon, and John K. Factor. The development and evaluation of the Earth Gravitational Model 2008 (EGM2008). *Journal of Geophysical Research: Solid Earth*, 117(B4):B04406, 2012. doi: 10.1029/2011JB008916.

[33] C. R. Gates. A simplified model of midcourse maneuver execution errors. Technical Report Technical Report 32-504 (NASA-CR-53032), Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, 1963. URL `https://ntrs.nasa.gov/citations/19640003365`.

[34] D. S. Lauretta, S. S. Balram-Knutson, E. Beshore, W. V. Boynton, C. Drouet d'Aubigny, D. N. DellaGiustina, H. L. Enos, D. R. Golish, C. W. Hergenrother, E. S. Howell, C. A. Bennett, E. T. Morton, M. C. Nolan, B. Rizk, H. L. Roper, A. E. Bartels, B. J. Bos, J. P. Dworkin, D. E. Highsmith, D. A. Lorenz, L. F. Lim, R. Mink, M. C. Moreau, J. A. Nuth, D. C. Reuter, A. A. Simon, E. B. Bierhaus, B. H. Bryan, R. Ballouz, O. S. Barnouin, R. P. Binzel, W. F. Bottke, V. E. Hamilton, K. J. Walsh, S. R. Chesley, P. R. Christensen, B. E. Clark, H. C. Connolly, M. K. Crombie, M. G. Daly, J. P. Emery, T. J. McCoy, J. W. McMahon, D. J. Scheeres, S. Messenger, K. Nakamura-Messenger, K. Righter, and S. A. Sandford. Osiris-rex: Sample return from asteroid (101955) bennu. *Space Science Reviews*, 212(1–2):925–984, 2017. doi: 10.1007/s11214-017-0405-1.

[35] J. T. Emmert, M. Jones, Jr., D. E. Siskind, D. P. Drob, J. M. Picone, M. H. Stevens, S. M. Bailey, S. Bender, P. F. Bernath, B. Funke, M. E. Hervig, and K. Pérot. NRLMSIS 2.1: An empirical model of nitric oxide incorporated into MSIS. *Journal of Geophysical Research: Space Physics*, 127(10):e2022JA030896, 2022. doi: 10.1029/2022JA030896.

[36] National Oceanic and Atmospheric Administration, National Aeronautics and Space Administration, and United States Air Force. U.S. standard atmosphere, 1976. Technical Report NOAA-S/T 76-1562 (NASA-TM-X-74335), U.S. Government Printing Office, Washington, DC, 1976. URL `https://ntrs.nasa.gov/citations/19770009539`.

[37] Daniel R. Wibben, Andrew Levine, Jason R. Russell, Anna T. Montgomery, James V. McAdams, Samantha Rieger, Peter G. Antreasian, and Kenneth M. Getzandanner. OSIRIS-REx earth return and entry: Targeting strategy and maneuver performance. In *46th Annual AAS Guidance, Navigation and Control Conference*, number AAS 24-058, Breckenridge, CO, 2024. URL `https://ntrs.nasa.gov/citations/20240000803`.

[38] Scott R. Francis, Mark A. Johnson, Eric Queen, and R. Anthony Williams. Entry, descent, and landing analysis for the OSIRIS-REx sample return capsule. In *46th Annual AAS Guidance, Navigation and Control Conference*, Breckenridge, CO, 2024. URL `https://ntrs.nasa.gov/citations/20240000629`.

[39] Astrobotic Technology. Peregrine mission one: Post-mission report. Technical report, Astrobotic Technology, Inc., Pittsburgh, PA, 2024. URL `https://www.astrobotic.com/wp-content/uploads/2024/08/PM1_Post-Mission-Report_Website.pdf`.

[40] R. Mitcheltree, R. Wilmoth, F. Cheatwood, G. Brauckmann, and A. Greene. Aerodynamics of stardust sample return capsule. In *15th Applied Aerodynamics Conference*. American Institute of Aeronautics and Astronautics, 1997. doi: 10.2514/6.1997-2304.

[41] J. M. Picone, A. E. Hedin, D. P. Drob, and A. C. Aikin. NRLMSISE-00 empirical model of the atmosphere: Statistical comparisons and scientific issues. *Journal of Geophysical Research: Space Physics*, 107(A12):SIA 15–1–SIA 15–16, 2002. doi: 10.1029/2002JA009430.

[42] James M. Russell, III, Martin G. Mlynczak, Larry L. Gordley, Joseph J. Tansock, Jr., and Roy W. Esplin. Overview of the SABER experiment and preliminary calibration results. In *Optical Spectroscopic Techniques and Instrumentation for Atmospheric and Space Research III*, volume 3756 of *Proceedings of SPIE*, pages 277–288. SPIE, 1999. doi: 10.1117/12.366382.

[43] Martine De Mazière, Anne M. Thompson, Michael J. Kurylo, Jeannette D. Wild, Germar Bernhard, Thomas Blumenstock, Geir O. Braathen, James W. Hannigan, Jean-Christopher Lambert, Thierry Leblanc, Thomas J. McGee, Gerald Nedoluha, Irina Petropavlovskikh, Gunther Seckmeyer, Paul C. Simon, Wolfgang Steinbrecht, and Susan E. Strahan. The Network for the Detection of Atmospheric Composition Change (NDACC): history, status and perspectives. *Atmospheric Chemistry and Physics*, 18(7):4935–4964, 2018. doi: 10.5194/acp-18-4935-2018.

[44] J. T. Emmert, D. P. Drob, J. M. Picone, D. E. Siskind, M. Jones, Jr., M. G. Mlynczak, P. F. Bernath, X. Chu, E. Doornbos, B. Funke, L. P. Goncharenko, M. E. Hervig, M. J. Schwartz, P. E. Sheese, F. Vargas, B. P. Williams, and T. Yuan. NRLMSIS 2.0: A whole-atmosphere empirical model of temperature and neutral species densities. *Earth and Space Science*, 8 (3):e2020EA001321, 2021. doi: 10.1029/2020EA001321.

## A Reproducing the numbers

Each table and figure comes from one command, run from the repository root with the services and data described in the runbook. Targets named `make` are in the repository. The scripts named "the report's" read the running API and are kept with the report's working files, outside the repository.

**Table 12:** Where each result comes from.

| Result | Command |
|---|---|
| Tables 3 to 5 | `make validate-trip` |
| Figure 4 | `make validate-trip`, drawn by the report's `make_fig_loops.py` |
| Shadow edges against SPICE | `test_eclipse_vs_spice.py`, printed by the report's `eclipse_check.py` |
| Table 8 and multi-year checks | `make hindcast-reforecast`, `make hindcast-hrrr` |
| Table 6 and the cosmic-ray and event checks | `make validate-radiation` |
| Table 7 and the footprint | `make validate-capsule` |
| Table 9 | `make validate-orbiters` |
| Rover line of sight | `make validate-viewshed` |
| Table 2, fig. 3 | the report's `export_data.py` and `make_figure.py` |
| Landing hindcast | `test_hindcast.py`; the report's `hindcast_surface.py` |
| Table 10 | the report's `measure_timing.py` and `sections.py --timing` |
| The compute queue | `make load-compute` |
| Figure 2 | `frontend/scripts/screens.ts` in a clean checkout, run by the report's `screens.sh` and cropped by `make_fig_screens.py` |
| Test counts | `make lint`, `make test-fast`, `npx vitest run`, `npx playwright test` |